
EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE MAJOR IN NETWORK AND INFORMATION SECURITY

De La Salle University – Manila, Philippines (SEPTEMBER 2022 – PRESENT)

- Current CGPA: 3.575 | Consistent and Consecutive First Dean's Lister

EXPERIENCE

IT AND WEB MAINTENANCE INTERN

AHEAD Education group – Quezon City, Philippines (MAY 2026 – JULY 2026)

- Designed and implemented a CRM platform that centralized student inquiry management, enrollment processing, payments, communications, and feedback collection. Created end-to-end UI/UX designs in **Figma** and translated them into a production-ready application. Developed secure role-based access controls, automated communication workflows, webhook integrations, and internal collaboration tools while contributing to database architecture and security planning. Leveraged **Next.js**, **TypeScript**, **Supabase**, and **Drizzle ORM** to build scalable staff-facing operations software.

PROJECTS AND PROGRAMS

MULTITHREADED OPERATING SYSTEM SCHEDULER SIMULATOR (2024)

- Developed a multithreaded operating system simulator in **C++** featuring a Round-Robin CPU scheduler, multi-core process execution, and virtual memory paging. Implemented thread-safe scheduling queues, process state management, page table and frame allocation mechanisms, and backing-store support while collecting CPU utilization and performance metrics. Leveraged C++ concurrency primitives, mutexes, condition variables, and atomic operations to simulate realistic operating system behavior.

CREATED A WEB APPLICATION AND APPLIED VULNERABILITY PATCHES (2025)

- Developed a full-stack restaurant review platform using **Node.js**, **Express**, **MongoDB**, and **Handlebars**, supporting role-based workflows for users, restaurant owners, managers, and administrators. Implemented secure authentication, account recovery, file uploads, review and rating systems, application logging, and extensive input validation. Enhanced platform security through password policies, login lockout protection, IP-based rate limiting, access controls, and vulnerability mitigation practices.

NETWORK PROTOCOL USING NAMED NETWORKING (2025)

- Designed and implemented a **Python**-based Named Data Networking (NDN) simulator that modeled content-centric routing, caching and distributed name resolution across interconnected nodes and name servers. Developed custom packet protocols, FIB/PIT/CS routing structures, UDP-based communication, multithreaded network processing, and performance analytics for measuring latency, throughput, and routing efficiency. Built a Tkinter visualization tool for inspecting network topology, routing tables, and simulations.

ACHIEVEMENTS, SKILLS, AND OTHER EXPERIENCES

CERTIFICATIONS, AND OTHER SKILLS

- Knowledgeable in Cybersecurity concepts and foundations and licensed with ISC2's Certificate in Cybersecurity (2025)
- Experienced in using Linux to perform security-related tasks such as Pentesting, Malware handling, Regex, and Digital forensics
- Experienced in using different programming languages such as: C, C++, Java, JavaScript, TypeScript, Python, Go, Assembly, etc.
- Experienced in UI/UX design using Figma, including wireframing, prototyping, user flows, and design systems
- Experienced in backend development using Node.js, Express.js, and Python, building RESTful APIs and authentication systems
- Experienced in designing and managing relational and NoSQL databases using SQL, PostgreSQL, MongoDB, Mongoose, and Supabase
- Fluent both verbally and written in English, Filipino, and Chinese

ORGANIZATIONAL EXPERIENCE

- Assistant Vice President and Candidate Vice President for ENGLICOM's Finance Committee (2024 – 2025)
- Assistant Team Head for ENGLICOM's flagship events such as: Chinese New Year, Medical Mission, and Nexus (2022 – 2024)
- Associate Vice President for La Salle Computer Society's Training and Development Committee (2024 – PRESENT)
- Project Head for "Encrypted: Mastering the Fight Against Data Breaches" and "GitHired: From Repos to Resumes" (2025)